PRESENTATION SCRIPT · 3 SPEAKERS

Docker-Based Virtual IoT Security Testbed

Botnet-attack simulation & anomaly-based intrusion-detection evaluation · Dept. of CSE (Cyber Security)

WHO SPEAKS WHAT

Speaker 1 — Aaruth V P · Slides 1–6 · ~4.5 min
The problem & the system

Speaker 2 — Devipriya F · Slides 7–11 · ~5 min
Detection + the live demo

Speaker 3 — Jeevabharathi S · Slides 12–17 · ~4.5 min
Comparison, real data & conclusion

SHAPE OF THE TALK

Total: ~13–14 minutes + Q&A

Flow: why it matters → how it works → the live demo → results → why it's better → close

The demo moment: Speaker 2 runs the live page on Slide 9 (`Run full experiment`), then returns to the slides.

Names are assigned in team order — swap freely.

FIVE RULES THAT MAKE IT LAND

- **Don't read the slides.** Each slide in the deck already has speaker notes — glance, then talk to the room.
- **Hand off by name** ("…now Devipriya will show how detection works") so the transitions feel planned.
- **One idea per slide.** Say the headline, give one sentence of detail, move on.
- **Pre-run the demo once** before you present; keep the offline page `demo/testbed-demo.html` open as backup.
- **Slow down on the results** (Slides 10–11) — that is the evidence the panel cares about.

Speaker 1 — Aaruth V P

Slides 1–6 · ~4.5 min

Opens the talk, states the problem, and gives the system overview.

1

Cover · introduce the team & the project

"Good morning. We are Aaruth, Devipriya and Jeevabharathi, final-year CSE Cyber Security. Our project is a Docker-based virtual IoT security testbed — a safe, reproducible way to simulate IoT botnet attacks and test an intrusion-detection system. I'll begin with why this matters."

2

The problem

"IoT devices are the number-one target for botnets. The Mirai botnet infected around 600,000 devices and took down major internet services. There are billions of these devices, many with weak default passwords. But testing defences on real hardware is expensive, risky and hard to repeat — so we need a safe, reproducible lab."

3

Aim & objectives

"Our aim is a containerized testbed that simulates attack traffic and evaluates an IDS — with a single command. We had five objectives: build a virtual IoT network, generate benign and attack traffic safely, detect anomalies with machine learning, measure the results live, and visualize everything on a dashboard."

4

Related work → the gap

"A lot of good work already exists — datasets like BoT-IoT and CICIoT2023, and detectors like Isolation Forest — but they are usually studied separately. Our contribution is to bring them together into one reproducible testbed."

5

System architecture

"Here is the system: eight Docker services on one isolated network. On the left, six IoT devices and the attack generator. In the middle, the MQTT broker, a message bus, and the Isolation-Forest IDS. On the right, the automation and the dashboard. Nothing touches the outside world, and it all starts with one command — docker compose up."

6

How it works — the pipeline

"It runs as one automated experiment in five stages: collect normal traffic, train the model, inject attacks, detect them, and report the results."

CUE — hand off: "Now Devipriya will explain how the detection actually works, and show it live."

→ passes to Speaker 2

Speaker 2 — Devipriya F

Slides 7–11 · ~5 min

Explains the detector and the attacks, runs the live demo, and presents the results.

7

Detection method

"Thank you, Aaruth. Detection uses an Isolation Forest. For every network flow we take features like bytes, packets, rates and destination port — plus two context features: how many ports and how many flows come from each source. It trains only on normal traffic, so it needs no attack labels, and it flags anything that looks unusual."

8

Attack signatures

"We reproduce four attack patterns — a port scan, a SYN flood, an MQTT flood, and a UDP flood. One important point: these are traffic signatures only. There is no real malware and no external hosts — this is exactly how public datasets like CICIoT2023 are built."

9

Live dashboard · RUN THE DEMO HERE

"This is our live dashboard — and instead of a screenshot, let me run it." `[open demo → click "Run full experiment"]`

"It collects normal traffic, trains the model, then injects the attacks — you can see attack traffic spike in red — and detects them in real time, filling the confusion matrix as it goes."

CUE — if the projector/Wi-Fi fails, open the offline file `demo/testbed-demo.html`; it needs no internet.

10

Results

"Here are the measured results: about 98% accuracy and 100% recall — it caught every attack — with a low false-positive rate. These numbers are computed live from the run, not hard-coded."

11

Detection quality — ROC & baseline

"And this is the key evidence. The ROC curve shows strong separation, and when we compare against a simple baseline detector on the same traffic, our method catches nearly twice as many attacks."

CUE — hand off: "Jeevabharathi will now cover how this compares to other testbeds and to real-world data."

→ passes to Speaker 3

Speaker 3 — Jeevabharathi S

Slides 12–17 · ~4.5 min

Covers the comparisons and real-data validation, then closes the talk.

12

Testbed comparison

“Thanks, Devipriya. Compared with virtual-machine and hardware testbeds, our Docker approach starts in about twelve seconds, is fully reproducible, and costs nothing extra — with the same strong isolation.”

13

Real-data evaluation

“We also run the exact same detector on real public datasets like CICIoT2023. So this is not only a simulation — the detector is validated on real-world attack traffic.”

14

What makes it stronger

“Compared with the earlier version of this project, every part is stronger: one-command Docker instead of manual setup, more device types, a better IDS, a live dashboard, automated experiments, and validation on real datasets.”

15

Conclusion

“In summary, we built a safe, reproducible IoT security testbed with a machine-learning IDS, live and honest metrics, a real-time dashboard, and a path to validate on real data.”

16

Future work

“Next steps include adding supervised models, using software-defined networking to automatically block detected attacks, more datasets, and testing on real IoT hardware.”

17

Thank you

“Thank you. We’re happy to take questions, and we can run the demo again. All of the code and the demo are on our GitHub.”

QUESTION & ANSWER — SPLIT & PREPARED

Who answers what

Aaruth (Speaker 1)

Architecture, Docker, reproducibility, project scope

Devipriya (Speaker 2)

The IDS, features, attacks, the live demo

Jeevabharathi (Speaker 3)

Results, comparisons, datasets, future work

“Is this real malware?”

No — it synthesizes attack-traffic **signatures** inside an isolated network, exactly like the BoT-IoT and CICIoT2023 datasets. No exploits, no self-propagation, no external hosts.

“Why isn’t the false-positive rate a fixed number?”

It’s a real detector — the false-positive rate is a **tunable operating point** (the contamination setting). We report what our run actually measured, not a fixed target.

“Why is recall almost 100%?”

In the testbed the attacks are strong statistical outliers, so they’re easy to separate. On raw real-data (Slide 13) recall is lower, and we report that honestly.

“How is it reproducible / does it run anywhere?”

Everything is in Docker. One command — `docker compose up` — recreates the exact environment on any Linux machine. We also ship a no-setup offline demo page.

“How does it scale?”

`docker compose up --scale dev-temp=20` adds twenty more devices instantly — one of the advantages of containers.

“Why Isolation Forest and not deep learning?”

It’s unsupervised, fast, and needs no attack labels — ideal for a live testbed. Comparing supervised and deep models is our stated future work.

Code & live demo: `github.com/Aaruth24/iot-botnet-testbed` · offline demo: `demo/testbed-demo.html`